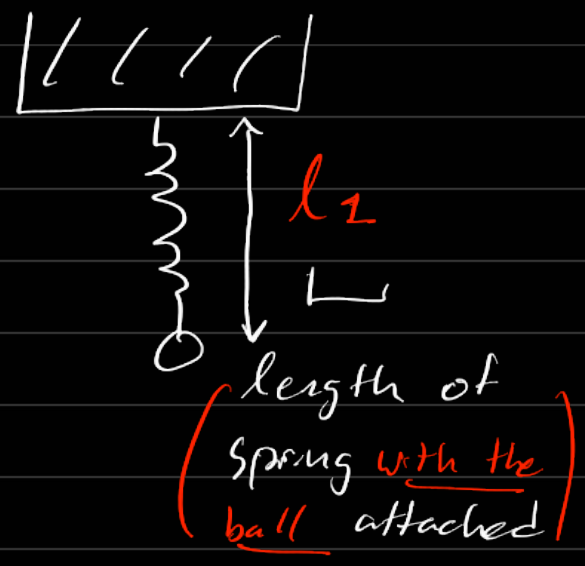
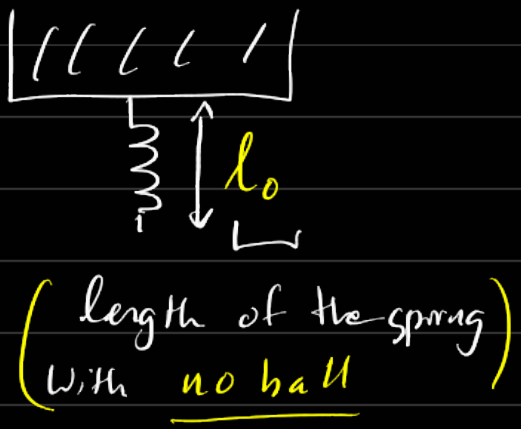


PS5 ②



• We originally wrote Hooke's law as

$$F_H = K \tilde{x}$$

$\tilde{x}$: Displacement from equilibrium.

However in this problem we are using the elastic modulus λ so the Force above F_H we write this now as

$$F_H = \lambda \frac{(l_1 - l_0)}{l_0}$$

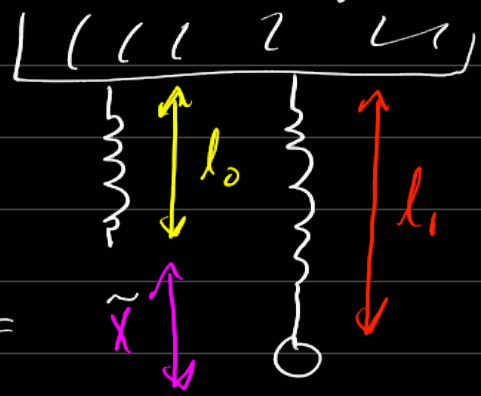
λ : elastic modulus

l_1 : length of spring with mass

l_0 : length of spring without mass (see picture)

→ Note the spring constant K and the modulus are related:

$$(l_1 - l_0) = \tilde{x} \text{ (length of displacement)}$$



$$l_1 - l_0 = \tilde{x}$$

$$(l_1 - l_0) = \tilde{x}$$

therefore $F_H = K x = \lambda \frac{(l_1 - l_0)}{l_0} = \frac{\lambda}{l_0} \tilde{x}$

so

$$K = \frac{\lambda}{l_0}$$

→ Going back to the problem:

When the spring is in equilibrium this means the sum

of the forces acting on it are zero

[L L L L]



$F_H \leftarrow$ force from spring

$F_g \leftarrow$ force from gravity

$$\sum F = 0 = F_H + F_g = 0$$



Note: F_g will be negative since this force points down.

$F_g = -mg$, thus

$$F_H + F_g = \frac{\lambda}{l_0}(l_1 - l_0) - mg = 0$$

Solve for l_1 :

$$l_1 = l_0 \left(1 + \frac{mg}{\lambda} \right)$$

⑥ Now, if the mass is in motion this means that the sum of forces is now: $ma = m \frac{d^2x}{dt^2}$

• Also, as we noted $l_1 - l_0 = x$ and so this will change over time:

$$F_H = \lambda \frac{x}{l_0}$$

$$\Rightarrow \sum F = F_H + F_g = \frac{\lambda}{l_0} \tilde{x} - mg = -m \frac{d^2 \tilde{x}}{dt^2}$$

and so we have

$$\frac{\lambda}{l_0} \tilde{x} - mg = -m \frac{d^2 \tilde{x}}{dt^2}$$

we can get rid of the $-mg$ factor by defining

$$X = \frac{\lambda}{l_0} \tilde{x} - mg$$

and so

$$\frac{d^2 X}{dt^2} = \frac{\lambda}{l_0} \frac{d^2 \tilde{x}}{dt^2} \Rightarrow \frac{d^2 \tilde{x}}{dt^2} = \frac{l_0}{\lambda} \frac{d^2 X}{dt^2}$$

plugging this back in:

$$X = -m \frac{l_0}{\lambda} \frac{d^2 X}{dt^2} = -\frac{ml_0}{\lambda} \ddot{X}$$

or

$$\ddot{X} + \frac{\lambda}{ml_0} X = 0$$

and

$$\omega^2 = \frac{\lambda}{ml_0}$$

(C) $\lambda = 1 \text{ N}$, $m = 100 \text{ g}$
 $l_0 = 10 \text{ cm}$, $g = 9.8 \text{ m/s}^2$

i) we use $l_1 = l_0 \left(1 + \frac{mg}{\lambda}\right)$
and just plug in

ii) Now, the true displacement is given by $\tilde{x}$ and not x since

$$l_1 - l_0 = \tilde{x}$$

thus $l_1 \equiv l(t) = l_0 + \tilde{x}(t)$

we defined x as

$$x = \frac{1}{l_0} \tilde{x} - mg$$

solving for $\tilde{x}$: $\tilde{x} = \frac{l_0}{\lambda} (x + mg)$

$$l(t) = l_0 + \frac{l_0}{\lambda} (x + mg)$$

$$l(t) = l_0 \left(1 + \frac{x + mg}{\lambda}\right)$$

Since $\ddot{\tilde{x}} + \omega^2 x = 0$
we can solve for x :

$$\lambda \left(\frac{l(t)}{l_0} - 1 \right) - mg = x$$

take $\frac{d^2}{dt^2}$: $\frac{d^2 \tilde{x}}{dt^2} = \frac{1}{l_0} \frac{d^2 l}{dt^2} = \frac{1}{l_0} \ddot{l}$

and replace

$$\ddot{\tilde{x}} + \omega^2 x = 0$$

↓ ↓

$$\left(\frac{1}{l_0} \right) \ddot{l} + \omega^2 \left[\lambda \left(\frac{l(t)}{l_0} - 1 \right) - mg \right] = 0$$

or

$$\frac{1}{l_0} \ddot{l} + \frac{\omega^2 \lambda}{l_0} l - \omega^2 \lambda - mg = 0$$

